

Jaewuk Moon

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WORK EXPERIENCE

Samsung Research (R&D center of Samsung Electronics) – Staff Engineer

Sep 2025 – present

The AI Model team has successfully developed Samsung Gauss (internal LLM) and on-device AI models for Galaxy series, including Sensor LLM for Galaxy Watch health monitoring.

□ Responsibilities

- Conducting LLM post-training research including supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) to improve Samsung Gauss model alignment and deployability.
- Developing **Sensor LLM** for Galaxy Watch health monitoring, applying time-series forecasting and self-supervised learning to extract health signals from noisy sensor data.
- Conducting **robotics** research on **vision-language-action (VLA)** models and **world action models**, coupling action-conditioned future prediction with policy learning to ground robot actions in physical dynamics.
- Leading independent research on **LLM-based ILI (Influenza-Like Illness) forecasting**, integrating multimodal inputs (numerical time-series + textual reports) with self-instructed fine-tuning to achieve explainable predictions.

Korea University Multimedia Information Laboratory – Ph.D. & Lab Manager

Mar 2019 – Aug 2025

The Multimedia Information Lab (MILab) at Korea University has successfully completed over 35 projects supported by major corporations and government agencies, published over 400 academic papers (including 100+ SCI-grade papers), registered 60+ patents, and focuses on cutting-edge AI and big data processing research in areas such as multimedia, predictive modeling, and healthcare technologies.

□ Responsibilities

- Led high-impact research initiatives, contributing to 18 international publications (8 first-author), 11 patent filings (10 granted), and multiple conference papers.
- As Lab Manager, streamlined research workflows, fostered interdisciplinary collaboration, and ensured successful project execution, significantly improving lab productivity.
- Managed administrative and performance tracking processes, supporting the BK21 research program and documenting progress.
- Assisted in the academic development of graduate students by managing seminars, assignments, and grading, ensuring the alignment of research objectives with academic milestones.
- Mentored junior researchers, guiding them to win 15 awards and publish 10 SCIE-indexed papers.

□ Selected Achievements

- **Infectious Disease Prediction Performance:** Achieved 200% of the project's publication target (6 papers published vs. 3 required) in COVID-19 projection research, collaborating with KISTI and Korea Disease Control and Prevention Agency.
- **Developed Custom LLM Solutions:** Designed and deployed custom LLM models for Samsung DS, integrating retrieval-augmented generation (RAG) to optimize industrial data workflows.
- **Advanced AI Techniques:** Enhanced prediction accuracy by 30% in applications like electricity usage forecasting, epidemic prediction, and anomaly detection, using Generative Adversarial Networks (GANs) and Model-Agnostic Meta-Learning (MAML) techniques.
- **B2B AI Agent Development:** Spearheaded the development of a multi-AI agent system for B2B solutions, enabling smart AI agent workflows for businesses like Incheon International Airport.
- **Explainable AI-Based Smart Energy Platform for Energy Optimization:** Applied Explainable AI (XAI) to optimize energy usage in industrial settings, saving over \$18,000 annually at Kiturami's Incheon factory.
- **Achieved Record-Breaking Publications:** Authored 8 SCIE first-author publications, the highest among lab members, establishing a strong record in academic publishing.

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EDUCATION

Korea University	Mar 2019 – Aug 2025
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Ph.D. in Electrical and Computer Engineering (GPA 4.11/4.5)

**Doctoral Dissertation: “An AI-based Influenza-like Illness Occurrence Forecasting Scheme based on LLM” (Advisor: Dr. Eenjun Hwang)*

Korea University	Mar 2015 – Feb 2019
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BE in **Electrical Engineering** (GPA 3.26/4.5)

INDUSTRY-ACADEMIC COOPERATIVE PROJECT

Energy Cloud Technology Development Project – Researcher	Jun 2019 – Dec 2023
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- Collaborated with the National Research Foundation of Korea (NRF) to develop a deep learning-based probabilistic model for energy consumption prediction and simulation, focusing on scalable solutions for microgrid systems.

Biodiversity Threat Management Technology Development Project – Researcher	Apr 2021 – Dec 2023
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- Developed a deep learning-based invasive species monitoring system in partnership with KEITI using acoustic data and AI techniques for environmental conservation.

AI-Based Building Architectural Design Solution Project – Researcher	Jul 2021 – Jun 2024
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- Led the design of an AI-powered architectural planning solution for the construction markets across multiple countries, including Singapore, Malaysia, and Vietnam, in collaboration with KIAT.

Infectious Disease Response Technology Development Project – Researcher	Apr 2019 – Dec 2022
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- Used deep learning to enhance predictive models for infectious diseases, including COVID-19, in collaboration with KHIDI (Korea Health Industry Development Institute).

Application of LLM solution and Developing AI Agent-based Services – Researcher	Jan 2024 – Jun 2025
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- Developed LLM-powered solutions, focusing on fine-tuning, prompt engineering, and RAG, to enhance the reliability and performance of AI systems.
- Created a self-evaluation framework to detect hallucinations in LLM-generated text, improving trust and accuracy in AI outputs.
- Designed hierarchical AI agents for optimizing AI workflows, as part of a broader initiative with LLOYDK.

EXTRACURRICULAR ACTIVITIES

Graduate Seminar – Teaching Assistant	Sep 2022 – Jun 2023
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- Organized guest speakers and coordinated schedules for Korea University's graduate seminar course.
- Managed student attendance, evaluated reports, and calculated final grades.

Korea University BK21 Data Analysis Fellowship – Project Manager	Jun 2021 – Aug 2021 / Dec 2021 – Feb 2022
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- Predicted power usage of major campus buildings to prevent outages and analyzed influencing factors using explainable AI (XAI) techniques.
- Enhanced the Korea University Data Hub system by leveraging LLM-based keyword extraction to recommend research-relevant papers to graduate students.

Yulin Technology Machine Learning Seminar – Instructor	Jun 2021 – Aug 2021
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- Delivered seminars on machine learning applications for a small manufacturing IT company, focusing on factory operations.
- Conducted lectures and coding workshops on data preprocessing, predictive modeling types, and their practical applications.

AWARDS & ACHIEVEMENTS

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- BK21 National Research Foundation of Korea Research Scholarship (2019 – 2024)
 - BK21 Data Analysis Fellowship (2021 – 2022)

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SKILLS & LANGUAGES

- Fluent in English (OPIc AL) and Korean
- Proficient in Microsoft Office Suite
 - Proficient in Python & Deep Learning Frameworks (TensorFlow & PyTorch)